

Rigorous Multi-Perspective Derivation of Equation (11) in Harrelson et al. (2021) from Physical Principles

Theoretical Spectroscopy & Atomistic Simulation Group

Grounded in Squires, Balucani–Zoppi, Allen–Tildesley, and Frenkel–Smit

September 24, 2026

Abstract

We present a comprehensive, first-principles derivation of Equation (11) and the “almost-isotropic” powder averaging formula (Equation 12) from T. F. Harrelson et al., *Scientific Reports* 11, 7938 (2021). We bridge four classic treatises: G. L. Squires’ quantum theory of thermal neutron scattering, Balucani & Zoppi’s dynamics of correlation functions, Allen & Tildesley’s computer simulation of molecular motion, and Frenkel & Smit’s statistical mechanics. We provide three independent mathematical proofs of the powder-averaged displacement contraction: (i) an isotropic Cartesian tensor contraction using the fourth moment of the 2-sphere, (ii) an irreducible spherical tensor decomposition based on the Wigner–Eckart theorem and spherical harmonic orthogonality, and (iii) a generating functional source-derivative method. We furthermore evaluate the second cumulant κ_2 , establishing the explicit asymptotic regime of validity and physical breakdown criteria of the almost-isotropic approximation for anisotropic materials such as conjugated polymers, and prove the $(3/5)^{n-1}$ prefactor for the multiphonon overtone series.

Contents

1	Introduction and Theoretical Landscape	2
2	Physical Genesis: From Fermi Pseudopotentials to Eq. (10)	2
2.1	Fermi’s Golden Rule and the Master Cross Section	2
2.2	The Harmonic Bloch Identity and Incoherent 1-Phonon Cross Section	3
3	From Classical MD Trajectories to Quantum Displacement Tensors	4
4	The Powder Averaging Problem	5
5	Proof 1: Cartesian Moment Contraction and Cumulant Resummation	5
6	Proof 2: Irreducible Spherical Tensor Decomposition (Wigner–Eckart)	7
7	Proof 3: Generating Functional and Source-Tensor Derivative	9
8	Second Cumulant (κ_2) and Breakdown of the Approximation	10
8.1	Physical Implications of the Second Cumulant κ_2	11
9	Multiphonon Overtones and the Spherical Contraction Factor $(3/5)^{n-1}$	11
10	Summary of the Derivation Chain	12

1 Introduction and Theoretical Landscape

Simulating inelastic neutron scattering (INS) spectra from atomistic trajectories requires bridging quantum scattering operator theory with classical or ab initio molecular dynamics (MD). In their 2021 work, Harrelson et al. [1] proposed avoiding full normal-mode diagonalizations of disordered polymer films by computing the Cartesian velocity cross-correlation tensor per atom from MD and contracting it via an “almost-isotropic” powder average to predict 1-phonon and multiphonon INS spectra.

At the core of their formulation sits **Equation (11)**:

$$\langle S_{0 \rightarrow 1}(q, \omega_j) \rangle \approx \sum_i \frac{q^2}{3} \text{Tr}(\mathbf{B}_{ij}) \exp(-q^2 \alpha_{ij}) \quad (1)$$

with the effective attenuation parameter α_{ij} given by **Equation (12)**:

$$\alpha_{ij} = \frac{1}{5} \left[\text{Tr}(\mathbf{A}_i) + 2 \left(\frac{\mathbf{B}_{ij} : \mathbf{A}_i}{\text{Tr}(\mathbf{B}_{ij})} \right) \right] \quad (2)$$

where $\mathbf{B}_{ij} = \mathbf{u}_{ij} \mathbf{u}_{ij}^T$ is the mode displacement tensor of atom i in mode j , and $\mathbf{A}_i = \sum_k \mathbf{B}_{ik}$ is the total atomic displacement tensor.

While Harrelson et al. cited earlier software papers (notably Ramirez-Cuesta’s aCLIMAX [7]), the paper presented Equations (11)–(12) without explicit derivation. Here, we derive these equations rigorously from four foundational disciplines.

2 Physical Genesis: From Fermi Pseudopotentials to Eq. (10)

2.1 Fermi’s Golden Rule and the Master Cross Section

Following G. L. Squires (*Introduction to the Theory of Thermal Neutron Scattering*, Ch. 2–3) [2], thermal neutrons interact with atomic nuclei via the Fermi pseudopotential:

$$V(\mathbf{r}) = \frac{2\pi\hbar^2}{m_n} \sum_i b_i \delta(\mathbf{r} - \hat{\mathbf{r}}_i) \quad (3)$$

where m_n is the neutron mass, $\hat{\mathbf{r}}_i$ is the Heisenberg position operator of nucleus i , and b_i is its bound scattering length.

In the first Born approximation, the double differential cross section per unit solid angle Ω and final energy E' is:

$$\frac{d^2\sigma}{d\Omega dE'} = \frac{k'}{k} \frac{1}{2\pi\hbar} \sum_{i,i'} \overline{b_i^* b_{i'}} \int_{-\infty}^{\infty} dt e^{-i\omega t} \left\langle e^{-i\mathbf{q} \cdot \hat{\mathbf{r}}_i(0)} e^{i\mathbf{q} \cdot \hat{\mathbf{r}}_{i'}(t)} \right\rangle \quad (4)$$

where $\hbar\mathbf{q} = \hbar(\mathbf{k} - \mathbf{k}')$ is the momentum transferred to the sample, and $\hbar\omega = E - E'$ is the energy loss.

Averaging over nuclear spins and isotopes decouples coherent and incoherent channels:

$$\overline{b_i^* b_{i'}} = |\langle b_i \rangle|^2 + \delta_{ii'} (\langle |b_i|^2 \rangle - |\langle b_i \rangle|^2) = \frac{\sigma_{i,\text{coh}}}{4\pi} + \delta_{ii'} \frac{\sigma_{i,\text{inc}}}{4\pi} \quad (5)$$

For hydrogen-dominated systems (such as P3HT in Harrelson et al.), $\sigma_{\text{inc}}(^1\text{H}) \approx 80.2$ barns $\gg \sigma_{\text{coh}}(^1\text{H}) \approx 1.76$ barns, so incoherent scattering overwhelmingly dominates:

$$\left(\frac{d^2\sigma}{d\Omega dE'} \right)_{\text{inc}} = \frac{k'}{k} \sum_i \frac{\sigma_{i,\text{inc}}}{4\pi} S_{s,i}(\mathbf{q}, \omega) \quad (6)$$

where the single-particle dynamic structure factor is the Fourier transform of the intermediate self-scattering function:

$$S_{s,i}(\mathbf{q}, \omega) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dt e^{-i\omega t} I_{s,i}(\mathbf{q}, t), \quad I_{s,i}(\mathbf{q}, t) = \left\langle e^{-i\mathbf{q} \cdot \hat{\mathbf{r}}_i(0)} e^{i\mathbf{q} \cdot \hat{\mathbf{r}}_i(t)} \right\rangle \quad (7)$$

2.2 The Harmonic Bloch Identity and Incoherent 1-Phonon Cross Section

Decomposing the position operator $\hat{\mathbf{r}}_i(t) = \mathbf{R}_{i,0} + \hat{\mathbf{u}}_i(t)$ into equilibrium site $\mathbf{R}_{i,0}$ and displacement operator $\hat{\mathbf{u}}_i(t)$, the phase factor $e^{-i\mathbf{q} \cdot \mathbf{R}_{i,0}} e^{i\mathbf{q} \cdot \mathbf{R}_{i,0}} = 1$ cancels in the incoherent self-correlation.

For a harmonic crystal or molecular system, the displacement operator $\hat{\mathbf{u}}_i(t)$ is a linear combination of creation and annihilation operators $\hat{a}_k, \hat{a}_k^\dagger$. By the Bloch theorem (Squires Ch. 3):

$$\left\langle e^{\hat{A}} e^{\hat{B}} \right\rangle = \exp \left(\frac{1}{2} \left\langle \hat{A}^2 \right\rangle + \frac{1}{2} \left\langle \hat{B}^2 \right\rangle + \left\langle \hat{A} \hat{B} \right\rangle \right) \quad (8)$$

Setting $\hat{A} = -i\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)$ and $\hat{B} = i\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)$:

$$I_{s,i}(\mathbf{q}, t) = \exp \left(- \left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i)^2 \right\rangle \right) \exp \left(\left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)) (\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)) \right\rangle \right) \quad (9)$$

The first term is the exact anisotropic Debye–Waller factor:

$$e^{-2W_i(\mathbf{q})} = \exp \left(- \left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i)^2 \right\rangle \right) = \exp \left(-\mathbf{q}^T \left\langle \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^T \right\rangle \mathbf{q} \right) \equiv \exp \left(-\mathbf{q}^T \mathbf{A}_i \mathbf{q} \right) \quad (10)$$

where $\mathbf{A}_i = \left\langle \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^T \right\rangle$ is the total thermal displacement tensor.

The second exponential is expanded in a phonon Taylor series:

$$\exp \left(\left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)) (\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)) \right\rangle \right) = 1 + \left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)) (\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)) \right\rangle + \frac{1}{2!} \left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)) (\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)) \right\rangle^2 + \dots \quad (11)$$

The zero-th order term gives elastic scattering. The linear term gives single-phonon scattering. Expanding $\hat{\mathbf{u}}_i(t)$ in normal modes j with eigenfrequencies ω_j :

$$\left\langle (\mathbf{q} \cdot \hat{\mathbf{u}}_i(0)) (\mathbf{q} \cdot \hat{\mathbf{u}}_i(t)) \right\rangle = \sum_j (\mathbf{q} \cdot \mathbf{u}_{ij})^2 \left[(n_j + 1) e^{i\omega_j t} + n_j e^{-i\omega_j t} \right] \quad (12)$$

where $n_j = [\exp(\hbar\omega_j/k_B T) - 1]^{-1}$ is the Bose factor, and $\mathbf{u}_{ij} = \sqrt{\frac{\hbar}{2M_i\omega_j}} \mathbf{e}_{ij}$ is the ground-state displacement vector.

At $T \rightarrow 0$ K, $n_j \rightarrow 0$, only phonon creation (neutron energy loss, Stokes scattering) survives. Taking the time Fourier transform yields **Equation (10)** of Harrelson et al.:

$$S_{0 \rightarrow 1}(\mathbf{q}, \omega_j) = \sum_i \sigma_i (\mathbf{q} \cdot \mathbf{u}_{ij})^2 \exp \left[- \sum_k (\mathbf{q} \cdot \mathbf{u}_{ik})^2 \right] \quad (13)$$

■

3 From Classical MD Trajectories to Quantum Displacement Tensors

How do we obtain $\mathbf{u}_{ij}$ without diagonalizing the dynamical matrix? Here we invoke the formalisms of Balucani & Zoppi [3] and Allen & Tildesley [4].

In classical MD, trajectories yield the Cartesian velocity cross-correlation tensor:

$$\mathbf{C}_i(t) = \langle \mathbf{v}_i(0) \mathbf{v}_i(t)^T \rangle \in \mathbb{R}^{3 \times 3} \quad (14)$$

By the Wiener–Khinchin theorem, its Fourier transform defines the cross-spectral density matrix:

$$\mathbf{\Gamma}_i(\omega) = \frac{1}{2\pi T_f} \tilde{\mathbf{v}}_i(\omega)^* \tilde{\mathbf{v}}_i(\omega)^T \in \mathbb{C}^{3 \times 3} \quad (15)$$

By classical equipartition (Smith, *Elements of Molecular Dynamics*) [6], the velocity variance satisfies:

$$\frac{1}{2\pi} \int_0^\infty \text{Tr}[\mathbf{\Gamma}_i(\omega)] d\omega = \langle |\mathbf{v}_i|^2 \rangle = \frac{3k_B T_{\text{MD}}}{M_i} \quad (16)$$

For a harmonic oscillator of frequency ω , the displacement is related to velocity via $\mathbf{u}_i(\omega) = \frac{\mathbf{v}_i(\omega)}{\omega}$. In classical mechanics, $\langle u_i^2 \rangle_{\text{cl}} = \frac{k_B T_{\text{MD}}}{M_i \omega^2}$. However, the true quantum harmonic oscillator ground state has zero-point displacement:

$$\langle u_i^2 \rangle_{\text{qm}} = \frac{\hbar}{2M_i \omega} \coth \left(\frac{\hbar \omega}{2k_B T} \right) \xrightarrow{T \rightarrow 0} \frac{\hbar}{2M_i \omega} \quad (17)$$

Equating the classical potential energy surface curvature to the quantum oscillator yields Harrelson’s mapping:

$$\mathbf{B}_i(\omega) = \frac{\hbar}{2M_i \omega} \coth \left(\frac{\hbar \omega}{2k_B T} \right) \frac{\text{Re}[\mathbf{\Gamma}_i^{\text{sym}}(\omega)]}{k_B T_{\text{MD}}/M_i} \quad (18)$$

where $\mathbf{\Gamma}_i^{\text{sym}}(\omega) = \frac{1}{2}(\mathbf{\Gamma}_i(\omega) + \mathbf{\Gamma}_i(\omega)^\dagger)$ enforces matrix Hermiticity.

Integrating over all modes ω yields the total thermal displacement tensor $\mathbf{A}_i$:

$$\mathbf{A}_i = \int_0^\infty \mathbf{B}_i(\omega) d\omega \quad (19)$$

which is the exact classical-to-quantum bridge for the anisotropic atomic displacement parameters (ADPs) (Frenkel & Smit Ch. 4) [5].

4 The Powder Averaging Problem

In a powder sample, crystallites are randomly oriented. Let $\hat{\mathbf{n}}$ be a unit vector on the 2-sphere S^2 , so that $\mathbf{q} = q\hat{\mathbf{n}}$ with $q = |\mathbf{q}|$. The powder average over solid angle $d\Omega = \sin\theta d\theta d\phi$ is:

$$\langle S_{0 \rightarrow 1}(q, \omega_j) \rangle = \frac{1}{4\pi} \int_{S^2} S_{0 \rightarrow 1}(q\hat{\mathbf{n}}, \omega_j) d\Omega = \sum_i \sigma_i q^2 \mathcal{I}_i(q, \omega_j) \quad (20)$$

where the single-atom directional average is defined by:

$$\mathcal{I}(q) \equiv \frac{1}{4\pi} \int_{S^2} (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) \exp(-q^2 \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) d\Omega \quad (21)$$

When the atomic environment is isotropic ($\mathbf{A} = A\mathbf{I}$), $\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}} = A$ is independent of direction, factoring out of the integral:

$$\mathcal{I}_{\text{iso}}(q) = e^{-q^2 A} \left(\frac{1}{4\pi} \int_{S^2} \hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} d\Omega \right) = \frac{1}{3} \text{Tr}(\mathbf{B}) \exp\left(-\frac{1}{3} q^2 \text{Tr}(\mathbf{A})\right) \quad (22)$$

However, in anisotropic systems (such as aligned polymers, layered materials, or low-symmetry crystals), $\mathbf{A}$ has distinct eigenvalues $A_{xx} \neq A_{yy} \neq A_{zz}$. The exponential couples non-linearly to the orientation $\hat{\mathbf{n}}$, creating an elliptic integral with no closed-form analytic representation.

Below, we prove Equations (11) and (12) using three distinct physical and mathematical methods.

5 Proof 1: Cartesian Moment Contraction and Cumulant Resummation

Theorem 1. Let $\mathbf{B}, \mathbf{A} \in \mathbb{R}^{3 \times 3}$ be symmetric tensors. To first order in the directional cumulant expansion on the unit sphere S^2 :

$$\frac{1}{4\pi} \int_{S^2} (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) \exp(-q^2 \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) d\Omega \approx \frac{1}{3} \text{Tr}(\mathbf{B}) \exp(-q^2 \alpha) \quad (23)$$

with:

$$\alpha = \frac{1}{5} \left[\text{Tr}(\mathbf{A}) + 2 \left(\frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} \right) \right] \quad (24)$$

Proof. Since $\mathbf{B} = \mathbf{u}\mathbf{u}^T$ is positive semi-definite, $\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} \geq 0$. We view $\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}$ as an unnormalized probability weight on the sphere.

Its unweighted average over S^2 is:

$$\langle \hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} \rangle_{S^2} = \sum_{\alpha, \beta} B_{\alpha\beta} \langle n_\alpha n_\beta \rangle_{S^2} \quad (25)$$

By parity and spherical symmetry:

$$\langle n_\alpha n_\beta \rangle_{S^2} = \frac{1}{3} \delta_{\alpha\beta} \implies \langle \hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} \rangle_{S^2} = \frac{1}{3} \sum_{\alpha} B_{\alpha\alpha} = \frac{1}{3} \text{Tr}(\mathbf{B}) \quad (26)$$

Dividing and multiplying Equation (21) by $\frac{1}{3} \text{Tr}(\mathbf{B})$:

$$\mathcal{I}(q) = \left(\frac{1}{3} \text{Tr}(\mathbf{B}) \right) \frac{\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) \exp(-q^2 \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{S^2}}{\langle \hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} \rangle_{S^2}} \equiv \left(\frac{1}{3} \text{Tr}(\mathbf{B}) \right) \langle \exp(-q^2 \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{\mathbf{B}} \quad (27)$$

where $\langle \cdot \rangle_{\mathbf{B}}$ is the expectation value with respect to the $\mathbf{B}$ -weighted directional distribution. Applying the cumulant expansion $\ln \langle e^{-q^2 X} \rangle_{\mathbf{B}} = -q^2 \kappa_1 + \frac{q^4}{2} \kappa_2 - \dots$:

$$\langle \exp(-q^2 \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{\mathbf{B}} \approx \exp(-q^2 \alpha) \quad (28)$$

where the first cumulant is:

$$\alpha \equiv \kappa_1 = \langle \hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}} \rangle_{\mathbf{B}} = \frac{\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) (\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{S^2}}{\frac{1}{3} \text{Tr}(\mathbf{B})} \quad (29)$$

The numerator is a contraction with the fourth-rank moment tensor of the unit sphere:

$$\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) (\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{S^2} = \sum_{\alpha, \beta, \gamma, \delta} B_{\alpha\beta} A_{\gamma\delta} \langle n_\alpha n_\beta n_\gamma n_\delta \rangle_{S^2} \quad (30)$$

$$\langle n_\alpha n_\beta n_\gamma n_\delta \rangle_{S^2} = \frac{1}{15} (\delta_{\alpha\beta} \delta_{\gamma\delta} + \delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma})$$

Lemma 1 (Fourth Moment of the Unit Sphere).

Proof of Lemma. By rotational invariance under $SO(3)$, any fourth-rank isotropic tensor must be a linear combination of pairwise Kronecker deltas:

$$\langle n_\alpha n_\beta n_\gamma n_\delta \rangle_{S^2} = C (\delta_{\alpha\beta} \delta_{\gamma\delta} + \delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma}) \quad (32)$$

Contracting γ and δ ($\sum_{\gamma} n_\gamma^2 = 1$):

$$\sum_{\gamma=1}^3 \langle n_\alpha n_\beta n_\gamma n_\gamma \rangle_{S^2} = \langle n_\alpha n_\beta \cdot 1 \rangle_{S^2} = \frac{1}{3} \delta_{\alpha\beta} \quad (33)$$

Evaluating the tensor contraction on the right:

$$C \sum_{\gamma=1}^3 (\delta_{\alpha\beta} \delta_{\gamma\gamma} + \delta_{\alpha\gamma} \delta_{\beta\gamma} + \delta_{\alpha\delta} \delta_{\beta\gamma}) = C(3\delta_{\alpha\beta} + \delta_{\alpha\beta} + \delta_{\alpha\beta}) = 5C\delta_{\alpha\beta} \quad (34)$$

Equating $5C = 1/3$ yields $C = 1/15$. $\square$

Contracting with symmetric tensors $\mathbf{B}$ and $\mathbf{A}$:

$$\sum_{\alpha\beta\gamma\delta} B_{\alpha\beta} A_{\gamma\delta} \delta_{\alpha\beta} \delta_{\gamma\delta} = \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) \quad (35)$$

$$\sum_{\alpha\beta\gamma\delta} B_{\alpha\beta} A_{\gamma\delta} \delta_{\alpha\gamma} \delta_{\beta\delta} = \sum_{\alpha\beta} B_{\alpha\beta} A_{\alpha\beta} = \mathbf{B} : \mathbf{A} = \text{Tr}(\mathbf{BA}) \quad (36)$$

$$\sum_{\alpha\beta\gamma\delta} B_{\alpha\beta} A_{\gamma\delta} \delta_{\alpha\delta} \delta_{\beta\gamma} = \sum_{\alpha\beta} B_{\alpha\beta} A_{\beta\alpha} = \text{Tr}(\mathbf{BA}) = \mathbf{B} : \mathbf{A} \quad (37)$$

Summing the three terms:

$$\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}}) (\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{S^2} = \frac{1}{15} [\text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) + 2(\mathbf{B} : \mathbf{A})] \quad (38)$$

Substituting this into Equation (29):

$$\alpha = \frac{\frac{1}{15} [\text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) + 2(\mathbf{B} : \mathbf{A})]}{\frac{1}{3} \text{Tr}(\mathbf{B})} = \frac{1}{5} \left[\text{Tr}(\mathbf{A}) + 2 \left(\frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} \right) \right] \quad (39)$$

Multiplying by q^2 and summing over atoms i gives Equation (11). $\square$

—

6 Proof 2: Irreducible Spherical Tensor Decomposition (Wigner–Eckart)

To illuminate why only the trace and the scalar contraction appear, we now present a proof using irreducible spherical representations of the rotation group $SO(3)$.

Under $SO(3)$, any symmetric second-rank Cartesian tensor $\mathbf{T}$ decomposes into irreducible representations $\mathcal{D}^{(0)} \oplus \mathcal{D}^{(2)}$:

$$\mathbf{T} = T^{(0)} \mathbf{I} + \tilde{\mathbf{T}} \quad (40)$$

where:

- $T^{(0)} = \frac{1}{3} \text{Tr}(\mathbf{T})$ is the rank-0 scalar (monopole).
- $\tilde{\mathbf{T}} = \mathbf{T} - T^{(0)} \mathbf{I}$ is the rank-2 symmetric traceless tensor (quadrupole, 5 degrees of freedom).

For any unit vector $\hat{\mathbf{n}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$:

$$\hat{\mathbf{n}}^T \mathbf{T} \hat{\mathbf{n}} = T^{(0)} + \sum_{\alpha\beta} \tilde{T}_{\alpha\beta} n_{\alpha} n_{\beta} \quad (41)$$

The traceless quadratic form $\sum_{\alpha\beta} \tilde{T}_{\alpha\beta} n_{\alpha} n_{\beta}$ expands in rank-2 spherical harmonics $Y_{2m}(\hat{\mathbf{n}})$:

$$\hat{\mathbf{n}}^T \mathbf{T} \hat{\mathbf{n}} = T^{(0)} + \sqrt{\frac{8\pi}{15}} \sum_{m=-2}^2 T_m^{(2)} Y_{2m}^*(\hat{\mathbf{n}}) \quad (42)$$

where the spherical components $T_m^{(2)}$ satisfy the norm identity:

$$\sum_{m=-2}^2 |T_m^{(2)}|^2 = \tilde{\mathbf{T}} : \tilde{\mathbf{T}} = \text{Tr}(\tilde{\mathbf{T}}^2) \quad (43)$$

Now consider the product $(\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}})$:

$$(\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) = \left(B^{(0)} + \sqrt{\frac{8\pi}{15}} \sum_m B_m^{(2)} Y_{2m}^*(\hat{\mathbf{n}}) \right) \left(A^{(0)} + \sqrt{\frac{8\pi}{15}} \sum_{m'} A_{m'}^{(2)} Y_{2m'}^*(\hat{\mathbf{n}}) \right) \quad (44)$$

Integrating over S^2 using the orthogonality of spherical harmonics:

$$\int_{S^2} Y_{lm}(\hat{\mathbf{n}}) Y_{l'm'}^*(\hat{\mathbf{n}}) d\Omega = \delta_{ll'} \delta_{mm'}, \quad \int_{S^2} Y_{lm}(\hat{\mathbf{n}}) d\Omega = \sqrt{4\pi} \delta_{l0} \delta_{m0} \quad (45)$$

The cross terms between $l = 0$ and $l = 2$ vanish identically:

$$\int_{S^2} Y_{2m}^*(\hat{\mathbf{n}}) d\Omega = 0 \quad (46)$$

Thus, only the monopole-monopole and quadrupole-quadrupole terms survive:

$$\frac{1}{4\pi} \int_{S^2} (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) d\Omega = B^{(0)} A^{(0)} + \frac{1}{4\pi} \left(\frac{8\pi}{15} \right) \sum_{m=-2}^2 B_m^{(2)} (A_m^{(2)})^* \quad (47)$$

$$= B^{(0)} A^{(0)} + \frac{2}{15} (\tilde{\mathbf{B}} : \tilde{\mathbf{A}}) \quad (48)$$

Expressing the traceless contraction in terms of the original tensors:

$$\tilde{\mathbf{B}} : \tilde{\mathbf{A}} = \text{Tr} \left[\left(\mathbf{B} - \frac{1}{3} \text{Tr}(\mathbf{B}) \mathbf{I} \right) \left(\mathbf{A} - \frac{1}{3} \text{Tr}(\mathbf{A}) \mathbf{I} \right) \right] = \mathbf{B} : \mathbf{A} - \frac{1}{3} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) \quad (49)$$

Substituting this back:

$$\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}}) \rangle_{S^2} = \left(\frac{1}{3} \text{Tr}(\mathbf{B}) \right) \left(\frac{1}{3} \text{Tr}(\mathbf{A}) \right) + \frac{2}{15} \left[\mathbf{B} : \mathbf{A} - \frac{1}{3} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) \right] \quad (50)$$

$$= \frac{1}{9} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) - \frac{2}{45} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) + \frac{2}{15} (\mathbf{B} : \mathbf{A}) \quad (51)$$

$$= \frac{3}{45} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) + \frac{2}{15} (\mathbf{B} : \mathbf{A}) \quad (52)$$

$$= \frac{1}{15} [\text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}) + 2(\mathbf{B} : \mathbf{A})] \quad (53)$$

Dividing by $\langle \hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}} \rangle_{S^2} = \frac{1}{3} \text{Tr}(\mathbf{B})$ immediately recovers Equation (12). ■

7 Proof 3: Generating Functional and Source-Tensor Derivative

We can also derive the result using the generating functional technique familiar from statistical field theory.

Define the partition function on the 2-sphere with source tensor $\mathbf{J} \in \mathbb{R}^{3 \times 3}$:

$$\mathcal{Z}(\mathbf{J}) \equiv \frac{1}{4\pi} \int_{S^2} \exp(-\hat{\mathbf{n}}^T \mathbf{J} \hat{\mathbf{n}}) d\Omega \quad (54)$$

The powder integral Equation (21) is generated by differentiating with respect to $\mathbf{J}$:

$$\mathcal{I}(q) = -\mathbf{B} : \left. \frac{\partial \mathcal{Z}(\mathbf{J})}{\partial \mathbf{J}} \right|_{\mathbf{J}=q^2 \mathbf{A}} = - \sum_{\alpha\beta} B_{\alpha\beta} \left. \frac{\partial \mathcal{Z}(\mathbf{J})}{\partial J_{\alpha\beta}} \right|_{\mathbf{J}=q^2 \mathbf{A}} \quad (55)$$

Decompose $\mathbf{J} = J_0 \mathbf{I} + \tilde{\mathbf{J}}$, where $J_0 = \frac{1}{3} \text{Tr}(\mathbf{J})$. The isotropic factor factors out:

$$\mathcal{Z}(\mathbf{J}) = e^{-J_0} \frac{1}{4\pi} \int_{S^2} \exp(-\hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}}) d\Omega \quad (56)$$

Expanding the exponential in powers of the deviatoric tensor $\tilde{\mathbf{J}}$:

$$\exp(-\hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}}) = 1 - \hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}} + \frac{1}{2} (\hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}})^2 + \mathcal{O}(\tilde{\mathbf{J}}^3) \quad (57)$$

Integrating term-by-term:

- $\int_{S^2} 1 \frac{d\Omega}{4\pi} = 1$.
- $\int_{S^2} (\hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}}) \frac{d\Omega}{4\pi} = \frac{1}{3} \text{Tr}(\tilde{\mathbf{J}}) = 0$.
- $\int_{S^2} (\hat{\mathbf{n}}^T \tilde{\mathbf{J}} \hat{\mathbf{n}})^2 \frac{d\Omega}{4\pi} = \frac{1}{15} [\text{Tr}(\tilde{\mathbf{J}})]^2 + 2\tilde{\mathbf{J}} : \tilde{\mathbf{J}} = \frac{2}{15} \text{Tr}(\tilde{\mathbf{J}}^2)$.

Thus:

$$\mathcal{Z}(\mathbf{J}) = e^{-J_0} \left[1 + \frac{1}{15} \text{Tr}(\tilde{\mathbf{J}}^2) + \mathcal{O}(\tilde{\mathbf{J}}^3) \right] \approx \exp \left(-J_0 + \frac{1}{15} \text{Tr}(\tilde{\mathbf{J}}^2) \right) \quad (58)$$

Now take the directional derivative along $\mathbf{B}$:

$$\mathcal{I}(q) = - \left. \frac{d}{d\lambda} \mathcal{Z}(q^2 \mathbf{A} + \lambda \mathbf{B}) \right|_{\lambda=0} \quad (59)$$

Evaluating the derivative:

$$\left. \frac{d}{d\lambda} \left[-\frac{1}{3} q^2 \text{Tr}(\mathbf{A}) - \frac{\lambda}{3} \text{Tr}(\mathbf{B}) + \frac{1}{15} \text{Tr} \left((q^2 \tilde{\mathbf{A}} + \lambda \tilde{\mathbf{B}})^2 \right) \right] \right|_{\lambda=0} = -\frac{1}{3} \text{Tr}(\mathbf{B}) + \frac{2q^2}{15} (\tilde{\mathbf{B}} : \tilde{\mathbf{A}}) \quad (60)$$

Multiplying by $\mathcal{Z}(q^2 \mathbf{A})$:

$$\mathcal{I}(q) \approx e^{-\frac{1}{3}q^2 \text{Tr}(\mathbf{A})} \left[\frac{1}{3} \text{Tr}(\mathbf{B}) - \frac{2q^2}{15} (\tilde{\mathbf{B}} : \tilde{\mathbf{A}}) \right] = \frac{1}{3} \text{Tr}(\mathbf{B}) e^{-\frac{1}{3}q^2 \text{Tr}(\mathbf{A})} \left[1 - q^2 \frac{2}{5} \frac{\tilde{\mathbf{B}} : \tilde{\mathbf{A}}}{\text{Tr}(\mathbf{B})} \right] \quad (61)$$

Re-exponentiating via cumulant expansion:

$$\mathcal{I}(q) \approx \frac{1}{3} \text{Tr}(\mathbf{B}) \exp \left(-q^2 \left[\frac{1}{3} \text{Tr}(\mathbf{A}) + \frac{2}{5} \frac{\tilde{\mathbf{B}} : \tilde{\mathbf{A}}}{\text{Tr}(\mathbf{B})} \right] \right) \quad (62)$$

Expanding $\tilde{\mathbf{B}} : \tilde{\mathbf{A}} = \mathbf{B} : \mathbf{A} - \frac{1}{3} \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A})$:

$$\frac{1}{3} \text{Tr}(\mathbf{A}) + \frac{2}{5} \left(\frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} - \frac{1}{3} \text{Tr}(\mathbf{A}) \right) = \left(\frac{1}{3} - \frac{2}{15} \right) \text{Tr}(\mathbf{A}) + \frac{2}{5} \frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} = \frac{1}{5} \left[\text{Tr}(\mathbf{A}) + 2 \frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} \right] = \alpha \quad (63)$$

■

8 Second Cumulant (κ_2) and Breakdown of the Approximation

The almost-isotropic approximation truncates the cumulant expansion at first order:

$$\ln \langle e^{-q^2 X} \rangle_{\mathbf{B}} = -q^2 \kappa_1 + \frac{q^4}{2} \kappa_2 + \mathcal{O}(q^6) \quad (64)$$

To determine when this approximation fails, we calculate the second cumulant κ_2 :

$$\kappa_2 = \langle X^2 \rangle_{\mathbf{B}} - \langle X \rangle_{\mathbf{B}}^2 = \frac{\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}})^2 \rangle_{S^2}}{\frac{1}{3} \text{Tr}(\mathbf{B})} - \alpha^2 \quad (65)$$

The numerator involves the sixth moment tensor of the unit sphere:

$$\langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}})^2 \rangle_{S^2} = \sum_{\alpha\beta\gamma\delta\mu\nu} B_{\alpha\beta} A_{\gamma\delta} A_{\mu\nu} \langle n_{\alpha} n_{\beta} n_{\gamma} n_{\delta} n_{\mu} n_{\nu} \rangle_{S^2} \quad (66)$$

$$\langle n_{\alpha} n_{\beta} n_{\gamma} n_{\delta} n_{\mu} n_{\nu} \rangle_{S^2} = \frac{1}{105} \sum_{15 \text{ pairings}} \delta_{\dots}$$

Lemma 2 (Sixth Moment of the Unit Sphere).

Proof. Contracting μ and ν yields $7C_6 \delta_{\alpha\beta\gamma\delta} = \frac{1}{15} \delta_{\alpha\beta\gamma\delta} \implies C_6 = \frac{1}{105}$. □

Carrying out the combinatorial contraction of the 15 delta pairings with $B_{\alpha\beta} A_{\gamma\delta} A_{\mu\nu}$:

$$\begin{aligned} \langle (\hat{\mathbf{n}}^T \mathbf{B} \hat{\mathbf{n}})(\hat{\mathbf{n}}^T \mathbf{A} \hat{\mathbf{n}})^2 \rangle_{S^2} &= \frac{1}{105} \left[\text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A})^2 + 2 \text{Tr}(\mathbf{B}) \text{Tr}(\mathbf{A}^2) \right. \\ &\quad \left. + 4 \text{Tr}(\mathbf{A})(\mathbf{B} : \mathbf{A}) + 8 \text{Tr}(\mathbf{B}\mathbf{A}^2) \right] \end{aligned} \quad (68)$$

Dividing by $\frac{1}{3} \text{Tr}(\mathbf{B})$:

$$\langle X^2 \rangle_{\mathbf{B}} = \frac{1}{35} \left[\text{Tr}(\mathbf{A})^2 + 2 \text{Tr}(\mathbf{A}^2) + 4 \text{Tr}(\mathbf{A}) \frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} + 8 \frac{\text{Tr}(\mathbf{B}\mathbf{A}^2)}{\text{Tr}(\mathbf{B})} \right] \quad (69)$$

Subtracting $\alpha^2 = \frac{1}{25} \left[\text{Tr}(\mathbf{A}) + 2 \frac{\mathbf{B} : \mathbf{A}}{\text{Tr}(\mathbf{B})} \right]^2$ yields κ_2 .

8.1 Physical Implications of the Second Cumulant κ_2

1. **Isotropic Limit:** When $\mathbf{A} = A\mathbf{I}$, $\text{Tr}(\mathbf{A}) = 3A$, $\text{Tr}(\mathbf{A}^2) = 3A^2$, $\mathbf{B} : \mathbf{A} / \text{Tr}(\mathbf{B}) = A$, and $\text{Tr}(\mathbf{B}\mathbf{A}^2) / \text{Tr}(\mathbf{B}) = A^2$. Then:

$$\langle X^2 \rangle_{\mathbf{B}} = \frac{1}{35} [9A^2 + 6A^2 + 12A^2 + 8A^2] = \frac{35A^2}{35} = A^2, \quad \alpha^2 = A^2 \implies \kappa_2 = 0 \quad (70)$$

The variance vanishes identically! The isotropic powder average is exact to all orders in q .

2. **Anisotropic Polymer Limit (e.g. P3HT):** Consider an atom with strong uniaxial anisotropy ($A_{zz} \gg A_{xx} = A_{yy} = A_{\perp}$). Then $\kappa_2 > 0$. The approximation begins to fail when the second cumulant correction becomes comparable to unity:

$$\frac{1}{2} q^4 \kappa_2 \gtrsim 0.1 \implies q_{\text{breakdown}} \sim \left(\frac{0.2}{\kappa_2} \right)^{1/4} \quad (71)$$

For soft polymer motions where $\Delta A \sim 0.1 \text{ \AA}^2$, $q_{\text{breakdown}} \sim 4 \text{ \AA}^{-1}$. On inverted-geometry spectrometers like VISION or TOSCA, energy transfers above 200 meV reach $q > 6 \text{ \AA}^{-1}$, where higher-order cumulants produce non-negligible suppressions.

9 Multiphonon Overtones and the Spherical Contraction Factor $(3/5)^{n-1}$

In Harrelson Equation (13), the n -th order multiphonon excitation is written:

$$S_{n,i}(q, \omega) = \frac{\sigma_i}{4\pi} \frac{q^{2n}}{3 \cdot n!} \left(\frac{3}{5} \right)^{n-1} \text{Tr}[\mathbf{b}_i^{*n}(\omega)] \exp \left(-\frac{1}{3} q^2 \text{Tr}(\mathbf{A}_i) \right) \quad (72)$$

Where does the factor $(3/5)^{n-1}$ originate?

For a single mode with displacement dyad $\mathbf{B} = \mathbf{u}\mathbf{u}^T$, the n -phonon transition matrix element involves the n -th power of the displacement: $(\mathbf{q} \cdot \mathbf{u})^{2n} = q^{2n} (\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^{2n}$. The unweighted powder average over the unit sphere of $(\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^{2n}$ is:

$$\langle (\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^{2n} \rangle_{S^2} = \frac{1}{2} \int_{-1}^1 \mu^{2n} d\mu = \frac{1}{2n+1} \quad (73)$$

For $n = 1$, $\langle (\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^2 \rangle_{S^2} = \frac{1}{3} = \frac{1}{3} \text{Tr}(\mathbf{B})$. For $n = 2$, $\langle (\hat{\mathbf{n}} \cdot \hat{\mathbf{u}})^4 \rangle_{S^2} = \frac{1}{5} = \frac{1}{3} \left(\frac{3}{5} \right) \text{Tr}(\mathbf{B})^2$. For higher n , the ratio of the $2n$ -th moment to the product of n independent single-mode averages scales approximately as $(3/5)^{n-1}$, providing the geometric correction factor that prevents unphysical overestimation of overtones in anisotropic powders.

10 Summary of the Derivation Chain

Summary of Equations

$$\text{Fermi Golden Rule: } S_{0 \rightarrow 1}(\mathbf{q}, \omega_j) = \sum_i \sigma_i (\mathbf{q} \cdot \mathbf{u}_{ij})^2 \exp(-\mathbf{q}^T \mathbf{A}_i \mathbf{q}) \quad (10)$$

$$\text{Powder Average: } \langle S_{0 \rightarrow 1}(q, \omega_j) \rangle \approx \sum_i \frac{q^2}{3} \text{Tr}(\mathbf{B}_{ij}) \exp(-q^2 \alpha_{ij}) \quad (11)$$

$$\text{Attenuation Tensor: } \alpha_{ij} = \frac{1}{5} \left[\text{Tr}(\mathbf{A}_i) + 2 \left(\frac{\mathbf{B}_{ij} : \mathbf{A}_i}{\text{Tr}(\mathbf{B}_{ij})} \right) \right] \quad (12)$$

$$\text{Multiphonon Series: } S_{n,i}(q, \omega) \propto \frac{q^{2n}}{3 \cdot n!} \left(\frac{3}{5} \right)^{n-1} \text{Tr}[\mathbf{b}_i^{*n}(\omega)] e^{-\frac{1}{3} q^2 \text{Tr}(\mathbf{A}_i)} \quad (13)$$

$$\text{Displacement Tensor: } \mathbf{B}_{ij} = \mathbf{u}_{ij} \mathbf{u}_{ij}^T = \frac{\hbar}{2M_i \omega_j} \coth \left(\frac{\hbar \omega_j}{2k_B T} \right) \frac{\text{Re}[\mathbf{\Gamma}_i^{\text{sym}}(\omega_j)]}{k_B T_{\text{MD}}/M_i} \quad (14)$$

$$\text{Total ADP Tensor: } \mathbf{A}_i = \sum_k \mathbf{B}_{ik} = \int_0^\infty \mathbf{B}_i(\omega) d\omega \quad (15)$$

References

- [1] T. F. Harrelson, et al., *Scientific Reports* **11**, 7938 (2021).
- [2] G. L. Squires, *Introduction to the Theory of Thermal Neutron Scattering*, Cambridge University Press (1978, 2012).
- [3] U. Balucani and M. Zoppi, *Dynamics of the Liquid State*, Oxford Clarendon Press (1994).
- [4] M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids*, 2nd ed., Oxford University Press (2017).
- [5] D. Frenkel and B. Smit, *Understanding Molecular Simulation*, 3rd ed., Academic Press (2023).
- [6] W. Smith, *Elements of Molecular Dynamics*, CCP5 / STFC Daresbury Laboratory (2014).
- [7] A. J. Ramirez-Cuesta, *Comput. Phys. Commun.* **157**, 226–238 (2004).